

A 1.1.3 CPU and GPU (HL only)



A 1.1.3 Explain the differences between the CPU and the GPU. (HL only)

The CPU and the GPU are answers to two different problems. The CPU is built to run anything: varied tasks, complex logic, decisions, one or a few things at a time at very low latency. The GPU is built to do one kind of work at massive scale: the same operation applied to thousands of data items at once. Every difference between them follows from that split, which is what makes this an Explain statement rather than a list to memorise. This page explains the two design philosophies and where each processor is used, compares their architecture, processing power, memory access and power efficiency, and explains how the two work together in practice.

Last Updated: 12 June 2026

Key Questions

- What problem is each processor designed to solve, and how does that shape where each is used?
- How do the CPU and GPU differ in core architecture, processing power, memory access and power efficiency, and why?
- When they work together, how is the work divided, how is data shared, and how is execution coordinated?

IB Command Term note

EXPLAIN: Give a detailed account including reasons or causes. Stating that the GPU has more cores will score less marks; explaining why (simple cores can be replicated by the thousand because they do not branch or decide) is what earns the marks.

WHAT TO INCLUDE: Design philosophies and usage scenarios. The syllabus includes four comparison dimensions (core architecture, processing power, memory access, power efficiency); and the three aspects of collaboration (task division, data sharing, coordinating execution).

Two design philosophies

The CPU is optimised for **serial** processing: handling a wide variety of tasks quickly, one or a few at a time. Its cores are sophisticated because the work demands it. Programs branch, take decisions, switch context and jump unpredictably between tasks. This is why the CPU runs the operating system, manages applications and handles user input. It is responsible for the overall management and execution of software.

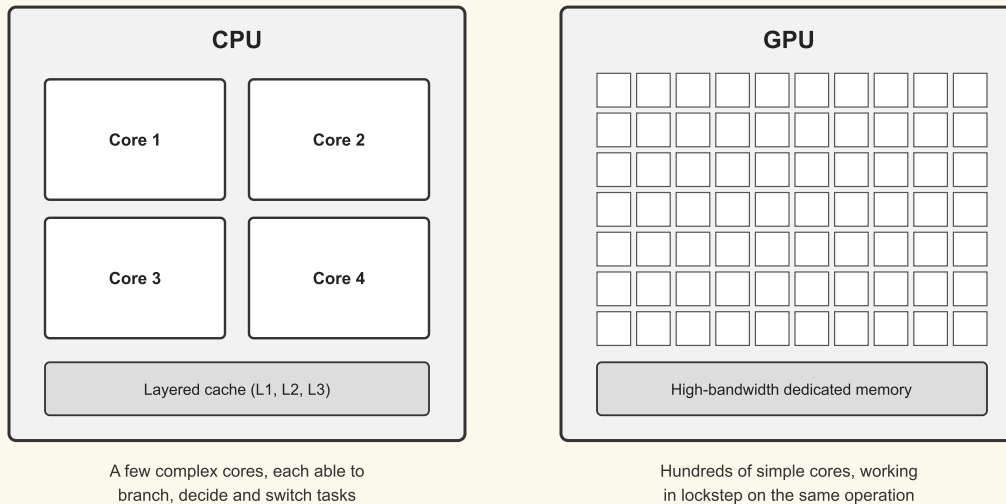
The GPU is optimised for **parallel** processing: performing the same operation on many data points simultaneously. Its cores are deliberately simple, because simplicity is what allows thousands of them to fit on one chip and run in lockstep. Originally built for graphics rendering, the GPU now carries any workload with the same shape: AI and machine learning, scientific simulation, large-scale data analysis.

Architecture, processing power, memory and power

The syllabus names four dimensions of comparison.

DIMENSION	CPU	GPU
Core architecture	A few sophisticated cores. Each can branch, take decisions and switch between tasks.	Hundreds or thousands of deliberately simple cores. They cannot branch or decide; they follow the same instruction in lockstep.
Processing power	High performance per core and very low latency on sequential work, branching and decision-making.	Enormous combined throughput when the work is parallel; weak when it is not, because the simple cores cannot handle varied sequential logic.
Memory access	Layered caches (L1, L2, L3) close to each core, because the CPU jumps unpredictably between tasks and needs recent data instantly.	Wide, high-bandwidth dedicated memory, because thousands of cores reach for data at the same moment and must all be fed without a bottleneck.
Power efficiency	More power-efficient per core; total consumption stays moderate because	Less efficient per core, but far more efficient per unit of parallel work

DIMENSION	CPU	GPU
	there are few cores.	completed: massive parallelism finishes the job in a fraction of the time.



Two design philosophies: low-latency sequential work (CPU) against massive parallel throughput (GPU)

Figure 1. Two design philosophies: low-latency sequential work (CPU) against massive parallel throughput (GPU).

COMMON EXAM MISTAKE

Answering 'which is more powerful' with a single winner. Generally speaking neither is more powerful. The CPU wins on sequential, branching work and the GPU wins on parallel work.

An Explain answer must give the reason, which is the workload shape, not the processor. A response that names one processor as “better” without conditions cannot reach the top of the markband.

Working together

In a real system the two processors work together and the syllabus names three aspects of the partnership.

- **Task division.** The CPU directs: it runs the program, decides which work is parallel, and assigns that work to the GPU. The GPU executes the compute-heavy portion and returns results.
- **Data sharing.** Data moves between the two over a high-speed connection, typically PCIe. Transfers cost time, so programmers batch work into large chunks to minimise the shuttling

back and forth.

- **Coordinating execution.** The CPU initiates GPU tasks, manages data dependencies and synchronises: it makes sure the GPU has finished before any dependent CPU work proceeds.

In gaming, the CPU handles game logic, AI and physics while the GPU renders the frames.

In machine learning, the CPU manages the training process and prepares the data while the GPU performs the matrix operations.

In video editing, the CPU runs the interface and encoding while the GPU accelerates effects and rendering.

Summary

SUMMARY REFERENCE

ASPECT	THE EXPLANATION TO GIVE
Design philosophy	CPU: serial, low-latency generalist for varied tasks and decisions. GPU: parallel throughput specialist for one operation over many data points.
Usage	CPU: operating system, applications, user input, overall management. GPU: rendering, AI and ML training, simulations, large-scale data work.
Four dimensions	Few complex cores vs many simple ones. Per-core power and low latency vs combined parallel throughput. Layered cache vs high-bandwidth memory. Per-core efficiency vs per-workload efficiency.
Collaboration	The CPU initiates, synchronises, and manages the master plan. The GPU takes the shared batches of data over PCIe and does the massive parallel heavy lifting.

Try it on paper

Explain why a GPU renders a video-game frame faster than a CPU could, even though each GPU core is simpler than a CPU core. [3 marks]

Model answer: Rendering a frame means performing the same calculations (lighting, texturing) on millions of pixels, and each pixel can be processed independently. [1] The GPU's hundreds or thousands of cores process huge numbers of pixels simultaneously, while a CPU's few cores would have to work through them in sequence. [1] The simplicity of GPU cores is the enabler, not a weakness: because they do not branch or decide, thousands fit on one chip and run in lockstep. [1]

Explain how the CPU and GPU divide and coordinate their work when training a machine learning model. [4 marks]

Model answer: The CPU manages the overall training process and prepares the data, because that work is varied and sequential. [1] It assigns the matrix operations, which are parallel and independent, to the GPU. [1] Data is transferred between them over a high-speed connection such as PCIe, batched into large chunks because transfers cost time. [1] The CPU synchronises execution: it waits for the GPU to complete each batch before proceeding with dependent steps such as updating the model. [1]

A 1.1.3 Quiz

1

Which statement BEST describes the fundamental difference in design philosophy between CPUs and GPUs?

- A. ☐ CPUs are faster than GPUs for all types of tasks.
- B. ☐ CPUs are designed for general-purpose tasks, while GPUs are designed for graphics-specific tasks.
- C. ☒ CPUs are optimized for serial processing, while GPUs are optimized for parallel processing.
- D. ☐ CPUs are larger and more complex than GPUs.



2

What is a key characteristic of CPU architecture?

- A. ☐ Specialised memory for parallel access.
- B. ☐ Simplified control flow for repetitive operations.
- C. ☐ A large number of simple cores.
- D. ☒ Fewer, more complex cores optimized for diverse tasks.

3

How does GPU memory access differ from CPU memory access?

- A. ☒ GPUs have larger, higher-bandwidth memory optimized for parallel access.
- B. ☐ GPUs do not have dedicated memory.
- C. ☐ GPUs have smaller, faster cache memory.
- D. ☐ GPUs have slower memory than CPUs.

4

Which type of processor is generally MORE power-efficient per core?

- A. ☐ Both are equally power-efficient.
- B. ☐ GPU
- C. ☐ Power efficiency depends on the specific model.
- D. ☒ CPU

5

In a typical gaming scenario, which task would MOST likely be handled by the CPU?

- A. ☐ Rendering 3D models of characters.
- B. ☐ Applying textures and lighting effects.
- C. ☐ Encoding the video output for display.
- D. ☒ Calculating physics and AI for non-player characters.

6

What is the primary role of the CPU in a CPU-GPU collaborative environment?

- A. ☐ To perform all the complex calculations.
- B. ☐ To only handle graphics rendering.
- C. ☐ To provide power to the GPU.
- D. ☒ To act as the master processor, delegating tasks and managing resources.

7

How do CPUs and GPUs typically share data?

- A. ☒ Through high-speed interconnects like PCIe.
- B. ☐ Through wireless communication channels.
- C. ☐ Through a shared memory pool.
- D. ☐ They do not share data directly.

8

Which of the following is NOT a common example of CPU-GPU collaboration?

- A. ☐ Training a machine learning model.
- B. ☐ Playing a graphics-intensive video game.
- C. ☐ Editing a video with special effects.
- D. ☒ Running a word processing application.

9

What is a potential advantage of using GPUs for scientific simulations?

- A. ☒ They can accelerate calculations due to their parallel processing capabilities.
- B. ☐ They consume less power than CPUs for all types of simulations.
- C. ☐ They can handle the complex branching and decision-making required in simulations.
- D. ☐ They are easier to program than CPUs for scientific tasks.

10

What is a common misconception about GPUs?

- A. ☐ GPUs are always faster than CPUs for any task.
- B. ☒ GPUs are only used for graphics rendering.
- C. ☐ GPUs consume more power than CPUs.
- D. ☐ GPUs have more cores than CPUs.

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